

Skin Lesion Classification Project Guide for the Team

Plain language project explanation and viva preparation / 28 September 2026

Our project is “Attention-Based Deep Learning Framework for Skin Lesion Classification with Explainable AI.” This guide starts from zero. We are teaching a computer program to look at a photograph of a mark on skin and suggest one of seven categories. That suggestion is a prediction. The trained program is called a model. This is a college research project, not a medical diagnosis. The later sections explain every special term, then describe what we actually tried and what the saved results mean.

1 The project in one minute

A skin lesion is a patch or mark that differs from the skin around it. A dermoscope is a tool that helps take a close-up picture of such a mark. Our task is to put each picture into one of seven named groups. A program that chooses a group is called a classifier. Its choice is only a prediction; a doctor would need much more information before making a diagnosis. Our aim is to study how well this picture-sorting task can be done and to show which picture areas influenced an answer.

We decided to build the final picture program from a specific family called EfficientNet, add a small focusing part called CBAM, and use a heatmap method called Grad CAM to inspect answers. These names are explained below. We also tried a different research program, PanDerm, to learn what other picture knowledge might add. PanDerm results are kept separate from the agreed final design. The final model choice and untouched final check have not happened yet.

2 Dataset and the seven classes

HAM10000 is a set of 10,015 close-up skin pictures with known category names. A category is also called a class. A small information file, called metadata, tells us each picture’s category and its lesion_id, the code identifying the particular skin mark. Some marks have more than one picture. The numbers of pictures in the seven groups are very uneven: the nv group has 6,705, while df has 115. We call that class imbalance. A program can appear good at sorting if it mostly recognizes the common group while missing rare ones.

Code	Simple meaning	Original images
nv	Melanocytic nevus, often called a mole	6,705
mel	Melanoma, a serious skin cancer	1,113
bkl	Benign keratosis like lesion	1,099
bcc	Basal cell carcinoma, a skin cancer	514
akiec	Actinic keratosis or related early cancer change	327
vasc	Vascular lesion, linked to blood vessels	142
df	Dermatofibroma, usually benign	115

The picture is the only information the main model sees when making a prediction. The metadata supplies the correct answer during learning and the lesion_id used to keep related pictures together. We briefly checked whether age, sex, and body location might help, but that small study did not give enough improvement to change the main image-only design.

3 How the learning system works

Term	Remember it this way
AI, machine learning, deep learning	AI is the broad goal of computer tasks that seem intelligent. Machine learning learns patterns from examples. Deep learning uses layered neural networks.
Model and CNN	A model is the learned set of rules and numbers. A CNN scans image patterns, building from edges and colors to richer visual features.
EfficientNet and B0, B1, B2	EfficientNet is a family of CNNs scaled in network width, depth, and input size. B0 is a smaller starting version; later B variants are larger and usually cost more memory and time. We actually trained B0, not B1 or B2.

Term	Remember it this way
Transfer learning and ImageNet	We began with EfficientNet weights trained on the large ImageNet image collection, then adjusted them on lesion images. It gives a useful starting representation.
CBAM	Convolutional Block Attention Module: a small learned unit placed after the backbone's final feature map. It reweights channels and spatial locations during prediction.
Channel and spatial attention	Channel attention asks which types of detected features matter. Spatial attention asks where on the feature map to emphasize them.
Grad CAM	A gradient based heatmap made for a selected prediction. It helps inspect which regions influenced the class score. It is not itself the learned CBAM module.

The words behind the computer program

Artificial intelligence or AI This is the broad idea of making a computer do tasks that appear intelligent. In our case, the task is sorting skin pictures into categories. AI does not mean the computer understands a disease as a doctor does.

Machine learning Instead of writing a separate rule for every mole shape and color, we show the computer many labeled pictures. It adjusts internal numbers so that its answers match the labels more often. Learning here means changing numbers through examples.

Deep learning and a neural network A neural network is a chain of many small calculation steps. "Deep" means it has many layers. Each layer passes a changed version of the picture information to the next one. Our project uses this kind of program.

CNN A convolutional neural network is a neural network designed for images. Early layers can detect simple edges or color changes; later layers combine these into more complex patterns. It is a picture-pattern finder, not a medical textbook.

Model, classifier, and feature A model is the trained program with its learned numbers. A classifier is a model used to choose a category. A feature is a useful pattern the model finds, such as an edge or texture. We do not hand-label every feature.

Prediction and probability A prediction is the category the model chooses. It also gives seven numerical scores that can be turned into probabilities adding to 100%. A high probability means the model favors a category; it is not proof the answer is correct.

EfficientNet This is the ready-made CNN family we used as the main picture-pattern finder. It was designed to use computing resources efficiently. We used its B0 version because it fit our available computer.

B0, B1, B2, and later versions These are differently sized members of the EfficientNet family. Later B numbers generally have a larger design and need more memory or computation. We trained B0; a paper or code project using B2 is a comparison, not a model we secretly trained.

Transfer learning and ImageNet pretrained Think of someone who already knows how to notice shapes in ordinary photographs, then learns skin pictures. ImageNet is a large general-image collection. We started from EfficientNet numbers learned there and adjusted them for the seven skin categories.

CBAM This is a small extra part attached to the EfficientNet picture features. It learns to give more or less importance to different patterns and positions while making an answer. It is part of the model during prediction.

Channel versus spatial attention A channel is one kind of pattern signal, such as a texture detector. Channel attention asks which signal types matter. Spatial attention asks which positions in the picture-like feature map matter. CBAM uses both in sequence.

Grad CAM and the difference from CBAM Grad CAM makes a colored heatmap after an answer to show which areas supported one category score. It is an inspection tool. CBAM changes feature strengths while the model works; Grad CAM helps us look back afterward. A bright heatmap area is not medical proof.

Now the complete path should make more sense: picture → prepare its size and colors → EfficientNet finds patterns → CBAM changes the importance of patterns and positions → the model gives seven category scores → one category is chosen. Grad CAM can then color parts of the picture linked to that answer. It does not increase the model's accuracy by itself.

A Simple Picture of EfficientNet and Attention

Imagine looking through a photo with many small magnifying glasses. Some notice edges, some notice color changes, and some notice texture. A CNN builds these observations step by step. EfficientNet is one ready-made way to organize that work without using an unnecessarily huge network. B0 is the version we trained; larger B numbers are other sizes in the same family. We started from ImageNet-trained numbers because they already help detect ordinary photo patterns. Training on skin pictures then adjusted those numbers for our seven categories.

CBAM is like a small focusing assistant inside the model. It first asks which kinds of detected patterns should count more, then asks where in the pattern map to pay more attention. It changes the calculation that leads to a prediction. Grad CAM does something different: after the model answers, it colors picture regions linked to one class score. The colored area gives us something to inspect, but it may include irrelevant background or camera artifacts. It is not proof that the program has the same understanding as a dermatologist.

4 Training and evaluation words

Term	Plain explanation
Train, validation, test	Training teaches the model. Validation helps choose settings and a checkpoint. The untouched test set is for final assessment after those choices are fixed.
Lesion level split and leakage	All photos of one lesion stay in one partition. Leakage occurs when information from training reaches an evaluation set, for example repeated lesion views on both sides.
Epoch, batch, batch size	An epoch is one pass through training references. A batch is one small group processed before an update. Batch size is how many images are in that group; our baseline used 64.
Learning rate and optimizer	Learning rate controls update size. AdamW is the optimizer that changes model weights using the loss.
Loss and checkpoint	Loss scores how wrong predictions are during learning. A checkpoint saves the model and training state; best.pt is selected by validation macro F1.
Early stopping and overfitting	Early stopping stops when validation stops improving. Overfitting means the model becomes good at seen examples but less good at new ones.
Augmentation and TTA	Training augmentation changes an image, for example by flipping it, to create variety. Test time augmentation, or TTA, averages predictions from different views at inference.
Oversampling, weighted loss, focal loss	Oversampling repeats minority training references. Weighted loss makes errors on rarer classes count more. Focal loss reduces emphasis on easy cases.
Ensembling	Average or combine predictions from several models or image views. It may correct different errors but requires more inference work.

How the program is taught and checked

Training and a training sample Training is the teaching process: show a picture, compare the model's answer with the known category, and adjust its internal numbers. One labeled picture used for this is a training sample.

Epoch One epoch means going through the whole list of training references once. Our original runs were allowed up to 20 epochs. More epochs are not always better because the model can begin memorizing what it has seen.

Batch and batch size We do not normally update the model after every single picture. A batch is a small group of pictures processed together. Batch size is how many are in that group; the main B0 runs used 64.

Loss Loss is a number that says how far the model's answer is from the known answer during training. Higher loss means it needs a stronger correction. The model tries to reduce loss over many batches.

Learning rate and optimizer An optimizer is the procedure that changes the model's internal numbers to reduce loss. Learning rate sets how large each change can be. Changes that are too large can be unstable; changes that are too small can learn slowly.

Validation and testing Validation pictures are set aside from training so we can check progress and choose settings. Test pictures are held aside even longer and should be used only after choosing the final method. A validation score is therefore a development score, not a final test result.

Checkpoint and early stopping A checkpoint is a saved copy of the model's learned numbers. We saved a "best" checkpoint when validation macro F1 improved. Early stopping means ending training when that measure stops improving instead of spending more time on unhelpful epochs.

Overfitting and augmentation Overfitting is like memorizing practice questions: performance on seen examples rises, but new examples remain hard. Augmentation changes training pictures in modest ways, such as flipping or rotating, to give more variety. It may help, but our stronger recipe did not improve validation results.

Weighted loss, oversampling, and focal loss These are three ways to respond to uneven class counts. Weighted loss makes mistakes on rare categories count more. Oversampling shows rare training pictures more often. Focal loss reduces attention to easy examples. We tested all three; none automatically solved the problem.

TTA and ensembling Test-time augmentation, or TTA, makes predictions from a picture and transformed views, then averages them. Ensembling averages predictions from several saved models or views. It can help when the different predictions make different mistakes.

Data leakage and lesion_id Leakage means the checking set gives an unfairly easy exam because related information was already seen during training. HAM10000 has several photos of some marks. We use lesion_id to put all photos of one mark in the same main train, validation, or test part.

In our main, stricter split, 7,009 pictures teach the model, 1,503 are for validation, and 1,503 are locked away for the final test. The same lesion_id never appears in two of these parts. In a separate exploratory split, only the development pictures were reshuffled by image. There, 563 lesion IDs appear in both training and validation, covering 596 validation pictures. Its results have a weaker independence guarantee, so we always label them "exploratory image-level validation." The original test pictures stayed excluded.

One Training Update Step by Step

Suppose one training picture is labeled melanoma. The model first produces seven category scores and favors the wrong group. A loss rule turns that mistake into a number. The optimizer uses that number to change the model's internal settings a little; learning rate limits the size of the change. It repeats this process over a batch of pictures, then over many batches in an epoch. After an epoch, validation pictures check whether the changes helped on pictures kept out of teaching. The best saved checkpoint can come from an earlier epoch, because extra practice can turn into memorization rather than improvement.

Weighted loss changes the loss rule so a mistake on a rare class matters more. Oversampling instead shows rare training pictures more often. Focal loss asks the model to spend relatively less effort on already easy pictures. We tested each idea, rather than assuming it would help. The results remind us that the training loss is a teaching signal, while validation accuracy and macro F1 are measures of how the learned model behaves on separate pictures.

5 Reading the scores

Metric	What it tells us
Accuracy	Correct predictions divided by all images. The common class can dominate it.

Metric	What it tells us
Precision	Of images predicted as a class, how many truly belong to that class?
Recall	Of real images of a class, how many did the model find?
F1	A balance of precision and recall for one class.
Macro F1	Average F1 across all seven classes, giving rare and common classes equal weight.
Confusion matrix	A table showing the real class against the predicted class, including which classes are confused.

How to read a result without being misled

Accuracy Imagine 100 pictures and 85 correct category choices. Accuracy is 85%. It is easy to understand, but the common nv category can dominate the count.

Precision If the model says “melanoma” ten times, precision asks how many of those ten pictures really are melanoma. It is about the trustworthiness of that particular predicted label.

Recall If there are ten real melanoma pictures, recall asks how many of those ten the model found. A model could have high precision but low recall if it almost never says melanoma.

F1 and macro F1 F1 combines precision and recall into one score for a category. Macro F1 calculates F1 for each of the seven categories, then takes a plain average. Rare categories therefore count as much as the common nv category.

Confusion matrix This is a grid of real categories against predicted categories. Its off-diagonal boxes show what the model mixed up, for example melanoma versus nevus. It tells more than a single percentage.

A number makes sense only with its category count, split, and measure. “90.75% accuracy on exploratory image-level validation” is accurate. “90.75% final test accuracy” would be false: the final test has not been run. Also, the best strict accuracy and best strict macro F1 came from two different combinations of models.

Why One High Percentage Can Be Misleading

Imagine 100 pictures: 80 are from the common nv group and only 20 are spread across the other six groups. A program that calls every picture nv would be 80% accurate, yet it would find none of the uncommon categories. Accuracy answers “How many pictures were right overall?” Macro F1 asks how well the program balanced precision and recall in each group, then averages those seven group scores. This is why our oversampling or prior-correction setting can gain accuracy and still lose macro F1. We need both measures and the confusion matrix.

Now imagine a student seeing two nearly identical photos of the same mark, one during practice and one on the exam. That exam is easier than a truly new mark. HAM10000 can contain more than one photo for one lesion, so we use lesion_id to keep related photos together in the main split. Our separate exploratory split intentionally does not preserve that rule. Its 90.75% is a real score on that development set, but it cannot be treated as the score for completely unseen lesions or the final locked test.

6 Other Research Checks in Plain Words

The next terms describe smaller checks we actually performed. Most reused saved models or saved picture patterns, so they did not require another full EfficientNet training run.

Frozen features We let a saved model turn each picture into pattern numbers and did not change that model. A smaller program tried to learn from those stored numbers.

Logistic regression and SVM These are two smaller pattern-sorting methods. We tested them on saved picture features; neither alone beat our leading image model on the strict split.

Metadata fusion This means combining picture predictions with extra information such as age, sex, or body location. Our cheap check did not improve enough to change the image-only plan.

MC dropout We let one model make several slightly varied answers and averaged them. The hope was a steadier answer, but it did not beat the existing leaders.

Crop check We tried a different way of cutting and resizing the picture only at checking time. It removed some visible area and hurt this saved model's result.

Prior correction We adjusted answer scores to account for changed class frequencies. It raised the number correct overall but favored the common class and reduced macro F1.

Picture size and PanDerm We averaged answers at 224 and 384 pixels. PanDerm is a different program pretrained on skin-related pictures. It helped as a research comparison but is outside our agreed final design.

7 Our project story

First we read and prepared the data

We reviewed HAM10000 and the base paper, counted the seven classes, and found large imbalance. We chose a fixed lesion separated split so related photographs would not appear in both training and validation. We built the image pipeline, EfficientNet B0 plus CBAM model, training and checkpoint code, evaluation outputs, and Grad CAM generation code.

Then we trained the core comparisons

Weighted cross entropy was our baseline because it helps rare classes without repeating their images. It reached 84.56% strict validation accuracy and 0.7569 macro F1. Training only oversampling reached 85.30% accuracy but 0.7321 macro F1; repeated minority references showed overfitting. Focal loss reached 83.77% and 0.7238, so it did not improve the leaders. These comparisons taught us that a higher accuracy can still mean less balanced performance.

Next we reused saved checkpoints

Horizontal flip TTA and probability averaging made modest changes. A weighted plus focal blend gave 0.7718 macro F1 at 84.36% accuracy on strict validation. Frozen feature classifiers, metadata fusion, MC dropout, and a crop diagnostic did not improve the best paired result. A prior correction raised strict accuracy to 85.89% but dropped macro F1 to 0.7227, mainly by favoring the common nevus class. We did not launch a large new training run for those cheap inference checks.

Then we tested why published percentages can look higher

An exploratory image level development split gave the weighted B0 model 86.23% accuracy and 0.7808 macro F1. Stronger augmentation alone scored lower on the same split: 85.30% and 0.7657. Averaging two saved B0 plus CBAM checkpoints at 224 and 384 pixel inference reached 88.16% and 0.8280. The same resolution idea did not improve strict validation. This taught us that inference gains can depend on the cohort and split.

Finally we made a separate research probe

PanDerm is a dermatology pretrained vision transformer, not an EfficientNet variant. A two epoch fine tuning pilot did not help on its own and was stopped. PanDerm features gave complementary predictions when combined with B0. The strongest strict accuracy ensemble reached 87.69%, while a different strict combination had the highest strict macro F1, 0.7837. The eight stream B0 plus PanDerm exploratory ensemble reached 90.75% with 0.8573 macro F1. It uses a split with shared lesions and is outside our approved final model family. It is a genuine validation result, but not a final or clinical accuracy claim.

What the major experiments taught us

The first weighted B0 run was a baseline: it asked whether rare classes could matter more without repeating their pictures. It gave 84.56% strict validation accuracy and 0.7569 macro F1. Repeating rare training pictures raised accuracy to 85.30%, but macro F1 fell to 0.7321, teaching us that one larger

number does not mean a more balanced model. Focal loss was another way to focus on difficult cases; its 83.77% accuracy and 0.7238 macro F1 did not improve the baseline.

Next, we reused saved models to avoid another long training job. Flip TTA and a small weighted-plus-focal average improved one balanced score. A prior adjustment pushed more answers toward the common class, raising accuracy but hurting macro F1. Checks using frozen image features, extra metadata, repeated dropout predictions, and a crop did not beat the leaders. These are useful findings because they tell us which inexpensive ideas did not justify a full new run.

A separate image-level split helped us understand why protocol matters, though it did not reach 93%. Stronger training flips and color changes made the training gap smaller but reduced validation performance. Averaging two saved EfficientNet models at two picture sizes improved that exploratory cohort to 88.16%. A different pretrained program, PanDerm, added useful complementary predictions and pushed an eight-view exploratory average to 90.75%. The strict lesion-separated check reached 87.69%, so 90.75% must stay tied to the exploratory split and the PanDerm component.

What Changed as the Project Progressed

We first prepared the picture collection and its labels, then built and trained the basic EfficientNet B0 plus CBAM system. The weighted-loss model was our reference. We tested repeated rare pictures and focal loss to see whether class balance would improve; neither beat the reference on both important measures. Next we tried cheaper checks with saved models: flipping pictures at prediction time, averaging model answers, using stored picture features, adding metadata, trying dropout, cropping, and adjusting class frequencies. Those checks taught us which ideas changed errors and which did not justify more training time.

Later we tested a different development split to understand protocol differences and used saved B0 models at two picture sizes. The four B0 streams gave 88.16% on exploratory validation. PanDerm, a separately pretrained research model, added different information and helped an eight-stream exploratory average reach 90.75%. The lesion-separated strict check reached 87.69%, so the 90.75% belongs only to its weaker split. The final chosen system must still follow EfficientNet, CBAM, and Grad CAM, and the locked test pictures have not yet been used.

8 Current results at a glance

Validation protocol and method	Accuracy	Macro F1
Strict: weighted B0 plus CBAM trained checkpoint	84.56%	0.7569
Strict: PanDerm plus B0 ensemble with highest accuracy	87.69%	0.7771
Strict: PanDerm plus B0 combination with highest macro F1	87.03%	0.7837
Exploratory shared lesion split: four B0 plus CBAM streams	88.16%	0.8280
Exploratory shared lesion split: eight B0 plus PanDerm streams	90.75%	0.8573

The highest numeric accuracy across all full validation methods is 90.75%. For the lesion separated strict protocol it is 87.69%. Among EfficientNet plus CBAM ensembles on the exploratory split it is 88.16%. These are not scores on the same evaluation protocol. The locked test set has no reported score.

9 The papers in simple language

Source	What they did and reported	What we learned
HAM10000 dataset paper [1]	Introduced the 10,015 image, seven category dataset.	Count classes and use lesion IDs so related images stay together.
Pardede base paper [2]	Compared EfficientNet B0 and MobileNetV3 Large with oversampling and Grad CAM; reported 98.86% and 98.88%.	EfficientNet is a sensible backbone and imbalance matters. Its 7,041 balanced test cases do not match an untouched 15% original image test, so percentages cannot be compared directly.

Source	What they did and reported	What we learned
Haque et al. [3]	Used much larger EfficientNetV2 L, attention, 384 pixel input and staged training; reported 91.15%.	Higher resolution and stronger training are research ideas; compute and split details differ from ours.
PanDerm [4]	Pretrained a dermatology focused transformer and released Base weights.	Specialized visual features may complement B0, but it is outside our final EfficientNet method.
Three category ensemble [5]	Reported 96% on a balanced selected HAM10000 subset.	A three category score cannot stand in for seven category performance.

For a viva, say that published percentages are real claims under their own methods. To compare fairly, check seven versus fewer classes, image versus lesion split, whether balancing touched validation or test, whether test was used to select a model, and whether the cohort was untouched. The base paper's exact data processing order cannot be proven from its public text; we should describe the inconsistency rather than accuse it of confirmed leakage.

For the base paper, our simple answer is: it used EfficientNet-B0 and balancing and reported 98.86%. We learned a useful architecture direction, but its 7,041 balanced test entries cannot be the untouched 15% of the 10,015 original HAM10000 images. We cannot prove exactly how the authors produced that table. Other high-paper results may use fewer categories, larger computers, image-level splits, or different data. We should explain those differences calmly instead of claiming that all papers made the same mistake.

How to Explain the Papers to the Guide

The HAM10000 paper gave us the picture collection and lesion identities. The Pardede base paper gave a closely related EfficientNet-B0 and balancing idea, but its reported 98.86% comes with a balanced 7,041-case test table, so we should not compare it as the same untouched test. The larger EfficientNetV2-L study gave a resolution and attention idea but used a much bigger model. PanDerm showed why skin-specific pretraining may be useful, even though it is outside our final design. A three-category ensemble reported 96%, but sorting three groups is different from sorting all seven. For every paper, answer four questions: what task, what method, what reported result, and how was its evaluation different from ours?

Sources for the paper summaries

- [1] Tschandl et al., HAM10000 dataset, 2018. <https://doi.org/10.1038/sdata.2018.161>
- [2] Pardede et al., Journal of Image and Graphics, 2026. <https://doi.org/10.18178/joig.14.4.551-566>
- [3] EfficientNetV2-L attention study, arXiv:2601.00964. <https://arxiv.org/html/2601.00964v1>
- [4] PanDerm paper and code. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12353815/> ; <https://github.com/SiyuanYan1/PanDerm>
- [5] Deep ensemble paper, Applied Sciences, 2024. <https://dSPACE.ewha.ac.kr/bitstream/2015.oak/269416/1/applsci-14-05599.pdf>

10 Likely viva questions and short answers

1. What is the goal?

Classify seven dermoscopic lesion categories and inspect predictions with Grad CAM.

2. Why HAM10000?

It is a public seven class dermoscopy dataset with image labels and lesion IDs.

3. What are the seven classes?

nv, mel, bkl, bcc, akiec, vasc, and df.

4. Is this a medical diagnosis?

No. It is a research classifier; clinical use needs further validation.

5. What is class imbalance?

Some classes have many more images; nv has 6,705 while df has 115.

6. Why EfficientNet?

It offers strong pretrained image features with moderate compute cost.

7. Why B0?

It is a practical starting variant for our local GPU; our core training used B0.

8. What is transfer learning?

Start with ImageNet pretrained weights and adapt them to lesion images.

9. Why CBAM?

It learns to emphasize useful feature channels and locations during prediction.

10. What is channel attention?

It weights types of visual features in the feature map.

11. What is spatial attention?

It weights positions in the feature map.

12. What is Grad CAM?

A gradient based heatmap for a chosen class prediction.

13. CBAM versus Grad CAM?

CBAM changes features inside the model; Grad CAM visualizes a prediction afterward.

14. What is a lesion level split?

All photographs of one lesion stay in one train, validation, or test partition.

15. What is leakage?

Information from training appears in evaluation, such as repeated lesion views.

16. What is an epoch?

One pass over the training references.

17. What is a checkpoint?

A saved model state; best.pt is chosen using validation macro F1.

18. Why use weighted loss?

It makes errors on rare classes count more without repeating validation images.

19. What happened with oversampling?

Accuracy rose to 85.30%, but macro F1 fell to 0.7321 on strict validation.

20. Why is accuracy insufficient?

The common class can dominate it while rare classes are missed.

21. What is macro F1?

The average of class F1 scores with every class counted equally.

22. What is TTA?

Average predictions from transformed views at inference.

23. Which method has the highest measured accuracy?

The exploratory eight stream B0 plus PanDerm ensemble: 90.75% validation, with shared lesions.

24. What is the best strict result?

87.69% accuracy from a PanDerm and B0 ensemble; a different configuration has 0.7837 macro F1.

25. Why do some papers report 96% to 99%?

Class sets, balancing, cohort selection and splitting often differ. Compare protocols before percentages.

26. Is the final result ready?

No. The original test set is untouched, and final model selection and Grad CAM review remain.

27. What is the approved final model family?

EfficientNet variants with CBAM and Grad CAM; PanDerm was a separate probe.

A safe viva answer to “What is your best result?” is: “Our highest measured full-validation accuracy is 90.75% on an exploratory image-level split using B0 plus PanDerm; it is not our final model or test result. The highest lesion-separated validation accuracy is 87.69% from another PanDerm/B0 combination. Our best EfficientNet plus CBAM exploratory ensemble is 88.16%. We still have to lock the approved final model and evaluate the untouched test once.” This wording may feel longer than saying one percentage, but it is honest and demonstrates that the team understands its own experiment.

11 How the Final System Will Work

After we settle the final EfficientNet plus CBAM settings using the development pictures, we will freeze them. For a new picture, the system will prepare its size and colors, use EfficientNet and CBAM to calculate seven category probabilities, show the chosen category, and optionally draw a Grad CAM heatmap for inspection. We will then run one final assessment on the locked test pictures and report accuracy, macro F1, class-by-class measures, and a confusion matrix. The heatmap and percentage will still be research evidence, not a clinical diagnosis.

12 What remains

We need to lock a final configuration consistent with the approved methodology, review Grad CAM examples carefully, run one final evaluation on the untouched test partition, and write the research paper with the exact protocol and limitations. Until then, all percentages in this guide are validation findings.

A simple walk through the final demo

Picture a new dermoscopic photograph that was never used in training. First, the saved preprocessing steps resize and normalize it in the same way as the development pictures. The chosen EfficientNet model reads patterns in the picture, and CBAM helps it give more attention to useful features. The last layer produces seven scores. We turn those scores into probabilities and display the category with the highest probability, alongside all seven probabilities. This is a research classification result; a doctor would still need clinical context and their own judgment.

For the same picture, Grad CAM makes a heatmap of regions that contributed to that category score. When showing it, say which class the heatmap explains. A warm patch can be a clue that the model used the lesion area, but it does not prove that the system reasoned like a dermatologist. If the heatmap highlights a corner, ruler, hair, or surrounding skin, the team should discuss that as a possible failure mode. Good examples and errors both belong in the final review.

How the team can present the evidence

Start the presentation with the seven-class task and the image collection. Explain why class imbalance and repeated views of a lesion make evaluation difficult. Then show one diagram: image, EfficientNet, CBAM, seven scores, and Grad CAM heatmap. After that, show a small comparison table with accuracy and macro F1 together. Label every row with its split and whether it uses the approved final architecture. A question about 90.75% should be answered with its exploratory image-level split and shared-lesion limitation before moving to the stricter result.

The final paper should tell the story in the same order: task, related papers, dataset, method, experiments, results, limits, and conclusion. The team can say that the highest measured accuracy in development is 90.75% under the exploratory protocol, while the lesion-separated result is lower. Those numbers show what has been measured so far. They do not promise the same score for a new patient, and they are not an untouched test score.